



Minia National University
Faculty Computers & Artificial Intelligence
Final Exam

Program: Artificial Intelligence

Course Title : Artificial Intelligence

Course Code : ACS 361

Lecturer : Dr. Alaa Zaki

Semester: Spring 2025

Date: 04 – 06 – 2025

Time: 2 Hours

Total Marks: 60 Marks

Model Answer

Answer the following question

Question 1: True or False [30 Marks]

- 1- Machine Learning (ML) is a subset of Artificial Intelligence (AI). ✓
- 2- Supervised learning requires labeled data for training. ✓
- 3- Unsupervised learning algorithms always need a target variable to predict. ✗
- 4- Reinforcement learning uses rewards and penalties to train models. ✓
- 5- A confusion matrix is used to evaluate regression models. ✗
- 6- Clustering groups similar data points together. ✓
- 7- Dimensionality reduction always leads to loss of information. ✗
- 8- Accuracy is the best metric for evaluating imbalanced datasets. ✗
- 9- Duplicate data entries should always be removed to avoid bias. ✓
- 10- One-hot encoding converts categorical variables into binary columns. ✗
- 11- Label encoding assigns arbitrary numbers to categories (e.g., "Pass"=1, "Fail"=2). ✓
- 12- Feature selection removes irrelevant features, while feature extraction creates new ones. ✓
- 13- In Machine learning model, humans write explicit rules. ✗
- 14- Traditional programming is more suitable for tasks with fixed, unchanging rules. ✓
- 15- Email spam detection (classifying emails as "spam" or "not spam") is an example of supervised learning. ✓
- 16- Predicting house prices based on features like location and size is a regression task in supervised learning. ✓
- 17- Autonomous driving (e.g., Tesla's Lane detection) uses only unsupervised learning. ✗
- 18- The goal of linear regression is to find θ_0 and θ_1 such that $h_{\theta}(x)$ exactly matches all training examples. ✗
- 19- If Cost Function = 0, the model perfectly predicts all training examples. ✓
- 20- Data augmentation can help reduce overfitting in deep learning models. ✓
- 21- The best-fit line minimizes the Cost Function between predictions $h_{\theta}(x)$ and true values y . ✓
- 22- One epoch means that the model has seen every training sample exactly once. ✓
- 23- The training set is a collection of data used in machine learning tasks. ✗
- 24- Fill in missing values and detect and eliminate causes of dataset exceptions called data cleansing. ✓
- 25- Overfitting occurs when the model or the algorithm does not fit the data well enough. ✓
- 26- Machine learning can be used when the Rules are complex or cannot be described. ✓

27- Normalization is used to increase the amount of training data by applying random transformations to existing data. ✗

28- Experience E refers only to labeled training data and cannot include unlabeled data. ✗

29- Support Vector Machines (SVM) can only be used for linear classification. ✗

30- Clustering algorithms like K-Means require labeled training data. ✗

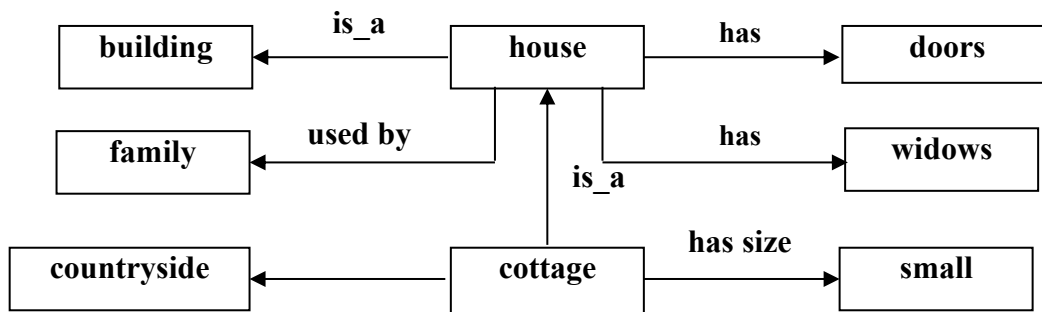
Question 2:

[10 Marks]

A) Represent the following English sentences using a *semantic network*:

[4 Marks]

A house is a building used by a family. All houses have doors and windows. A cottage is a kind of house that is small and in the countryside.



B) Translate the following sentences into predicate calculus expressions:

[6 Marks]

1. All children like candies.
2. If a child eats the food that he likes he is happy.
3. Tom is a child.
4. Toffee is candy.
5. Toffee is food.
6. Tom eats Toffee.

1. $\forall X \forall Y (\text{child}(X) \wedge \text{candy}(Y) \rightarrow \text{like}(X, Y))$

2. $\forall X \forall Y ((\text{child}(X) \wedge \text{food}(Y) \wedge \text{like}(X, Y) \wedge \text{eat}(X, Y) \rightarrow \text{happy}(X))$

3. $\text{child}(\text{tom})$

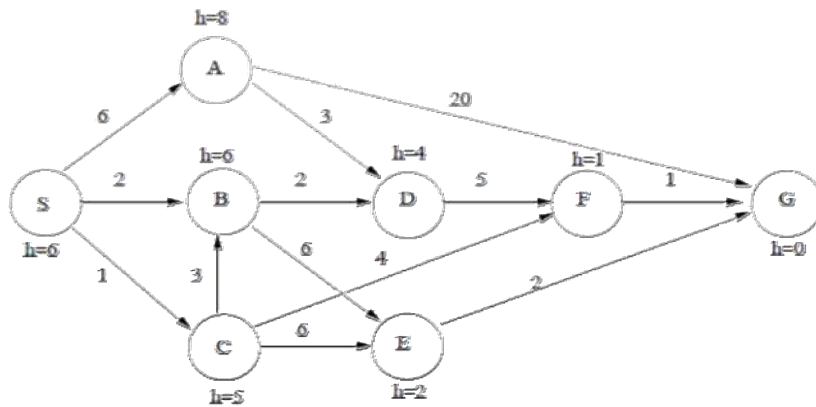
4. $\text{candy}(\text{toffee})$

5. $\text{food}(\text{toffee})$

6. $\text{eat}(\text{tom}, \text{toffee})$

Question 3: [10 Marks]

Consider the search problem below with start state S and goal state G. The transition costs are next to the edges, and the heuristic values are next to the states.



What is the final path for this search If we use A* search

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#	S	Frontier	Expand
0	S	S	
1	C	(A,F=14), (B,F=8), (C,F=6)	S
2	F	(A,14), (B,8,10), (E,8), (F,6)	S, C
3	G	(A,14), (B,8,10), (E,8), (G,6)	S, C, F
Final Path is S – C – F – G, Cost = 6			

Question 4: [10 Marks]

For the following training set of a small sample of different students' performances . Here each row is one training example. Recall that in linear regression, our hypothesis is $h_0(x) = \theta_0 + \theta_1 x$, and the cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^i) - y^i)^2$$

answer the following:

1-what is the value of m?

M = 5

2-What is $J(2,3)$?

$$\begin{aligned}
 J(2,3) &= \frac{1}{10} ((8 - 4)^2 + (11 - 2)^2 + (5 - 3)^2 + (17 - 4)^2 + (14 - 2)^2) \\
 &= \frac{1}{10} ((4)^2 + (9)^2 + (2)^2 + (13)^2 + (12)^2) \\
 &= \frac{1}{10} (16 + 81 + 4 + 169 + 144) = 41.4
 \end{aligned}$$

3-Suppose we set $\theta_0 = -3$, $\theta_1 = 2.5$. What is $h_0(4)$? $= -3 + 2.5 \times 4 = 7$

X	Y	h
2	4	$2+3*2=8$
3	2	$2+3*3=11$
1	3	$2+3*1=5$
5	4	$2+3*5=17$
4	2	$2+3*4=14$

With My Best Wishes